

Expressive Robotic Painting Through Imitation Learning: Learning Artistic Brush Stroke Styles from Human Demonstrations

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Abstract

Robotic drawing systems are commonly associated with precision, repeatability, and industrial control. This project investigates whether a simulated robotic manipulator can learn expressive artistic brush stroke movements from human demonstrations. A Unity 6 robotic painting environment was developed using Unity ML-Agents, consisting of a multi-joint robotic arm, a virtual canvas, a brush end effector, a demonstration recording pipeline, Behaviour Cloning training, ONNX model export, and a quantitative evaluation system. Three artistic motion styles were studied: smooth circular motion, spiral growth motion, and organic floral motion. Separate Behaviour Cloning policies were trained for each style and evaluated using mean trajectory error, path similarity, reproduction accuracy, completion success rate, path length, direction change, and curvature variance. The results show that Style A achieved the lowest mean trajectory error (0.01684), Style B produced moderate error (0.04317), and Style C produced the highest error (0.08526). These findings support the hypothesis that trajectory complexity negatively affects Behaviour Cloning performance in expressive robotic painting.

CCS Concepts

• **Computing methodologies** → **Imitation learning**; **Machine learning**; • **Computer systems organization** → *Robotics*; • **Applied computing** → *Fine arts*.

Keywords

Imitation Learning, Behaviour Cloning, Unity ML-Agents, Robotic Painting, Robotic Creativity, Simulation

1 Introduction

Robotic drawing and painting systems often emphasise industrial accuracy, path planning, and repeatable execution. Artistic mark-making, however, is not only defined by spatial precision. It also involves movement quality, rhythm, curvature, and stylistic variation. This project explores whether imitation learning can transfer expressive brush stroke movements from a human demonstrator to a simulated robotic arm.

The research question is: can Behaviour Cloning enable a robotic manipulator to reproduce different expressive artistic motion styles from human demonstrations? This directly follows Option B in the Applied Machine Learning project brief, which allows imitation learning for expressive robot movement in a creative or artistic domain. The hypothesis is that simpler motion styles will be reproduced more accurately than complex organic styles because

Behaviour Cloning is sensitive to trajectory complexity and compounding prediction error.

2 Related Work

Robot learning from demonstration provides a foundation for this project because it allows robotic skills to be specified through examples rather than hand-coded control rules. Argall et al. describe learning from demonstration as a practical approach when expert behaviour can be shown more easily than it can be analytically programmed. This is directly relevant to expressive painting: the desired behaviour is not only to reach a target point, but to reproduce a movement style involving timing, curvature, and visual continuity. For that reason, demonstration quality becomes part of the creative specification of the system.

Behaviour Cloning is one of the simplest imitation learning methods. It treats expert demonstrations as supervised state-action pairs and trains a policy to predict the demonstrated action from the current observation. Compared with reinforcement learning, Behaviour Cloning avoids the need to design a reward function for aesthetic or expressive quality. This is a major advantage in the present project, because “good” brush movement is difficult to reduce to a single scalar reward. Reinforcement learning methods such as PPO can be powerful when rewards are well specified and exploration is feasible, but expressive painting creates a sparse and subjective objective: a policy might receive high reward for reaching a geometric point while still producing visually poor strokes. Behaviour Cloning therefore offers a better fit for the project brief’s imitation learning option.

The limitation of Behaviour Cloning is that it can suffer from covariate shift. During inference, small prediction errors may move the robot into states that were not present in the demonstrations, causing further errors to accumulate. DAgger, proposed by Ross et al., addresses this problem by iteratively collecting corrective expert actions under the learner’s own state distribution. DAgger was not implemented here because the project prioritised a complete, stable Unity ML-Agents pipeline within the submission timescale, but it provides an important reference point for future improvement.

Unity ML-Agents was selected as the implementation framework because it supports robotic simulation, continuous control, demonstration recording, Behaviour Cloning training, and ONNX export within a single workflow. This practical integration made it possible to connect a simulated manipulator, demonstration data, trained policies, visual outputs, and quantitative evaluation in one reproducible project structure.

3 Creative Brief

The creative objective is to teach a simulated robotic arm to perform expressive painting gestures rather than industrially precise drawing. The final visual output is a set of brush stroke trajectories on a virtual canvas. The project uses three target styles: a layered circular ink movement, a spiral growth movement, and an organic floral/butterfly-inspired movement. These styles were chosen because they are visually interpretable, easy to compare, and represent increasing movement complexity. The demonstration dataset consists of recorded Unity ML-Agents demonstration files for policy training and CSV trajectory logs for quantitative evaluation.

The creative process involved designing a robot-canvas interaction in Unity, recording demonstrations, training Behaviour Cloning models, and comparing the generated robot paths with the demonstrated paths. The project frames creativity as expressive movement imitation: the robot is not judged only by exact endpoint accuracy, but by how closely it reproduces the dynamic structure of the demonstrated brush stroke.

4 Methodology

The project formulates robotic painting as an imitation learning problem, matching the brief’s Option B route. A user controls a simulated robotic arm and produces brush strokes on a virtual canvas. During demonstration, observation-action pairs are recorded for ML-Agents Behaviour Cloning, while brush trajectory coordinates are saved as CSV files for quantitative evaluation. Unlike the reinforcement learning option, this project does not define a reward function; the target behaviour is specified through expert demonstrations and learned as supervised state-action prediction.

Three styles were evaluated: Style A is a smooth circular motion, Style B is a spiral growth motion, and Style C is an organic floral motion. These styles represent increasing trajectory complexity, allowing the experiment to compare imitation performance across regular, time-varying, and organic paths.

5 Technical Implementation

The simulation was implemented in Unity 6. The system contains a robotic arm with base, shoulder, elbow, and wrist joints. The brush tip acts as the end effector and draws on a virtual canvas when it is close to the canvas plane. The visible drawing is produced by appending projected brush positions to rendered stroke geometry. Figure 1 shows how the architecture supports imitation learning: human demonstrations are captured as robot state-action behaviour, converted into training data, deployed as ONNX policies, and then evaluated against reproduced trajectories.

The ML-Agents agent uses continuous actions to control joint motion. Observations include joint state, brush position, brush velocity, and episode progress. Demonstration data is recorded using Unity ML-Agents demonstration files, and trajectory data is recorded separately in CSV format.

Behaviour Cloning configurations are provided for each style. Separate models are trained for Style A, Style B, and Style C, then exported as ONNX policies and assigned back into the Unity agent for inference. Separate models were used because each style has a different movement distribution: the circular policy learns periodic

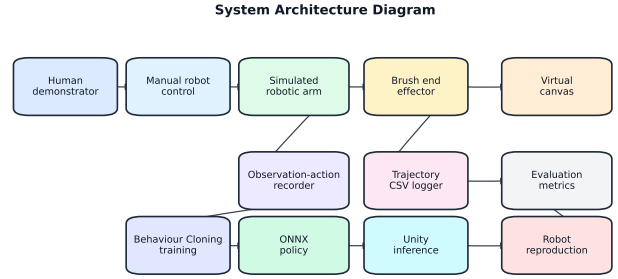


Figure 1: System architecture of the simulated robotic painting platform. The human demonstrator controls a virtual robotic arm, producing brush trajectories that are stored for Behaviour Cloning and later compared with trained robot reproductions.

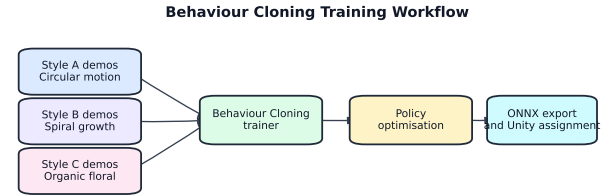


Figure 2: Behaviour Cloning training workflow. Separate models are trained for circular, spiral, and floral motion styles, with an optional combined model trained using all styles.

motion, the spiral policy learns changing radius over time, and the floral policy learns frequent local direction changes. This avoids forcing one small policy to learn style switching before the individual imitation problem has been evaluated. Figure 2 summarises this training and deployment workflow.

The training environment includes a Docker setup and helper scripts for running ML-Agents training. The complete repository contains the Unity project, Python training configuration, ONNX models, evaluation data, and documentation. The complete GitHub repository is available at:

<https://github.com/Jingtong112/ExpressiveRoboticPainting>.

6 Evaluation

Evaluation compares demonstrated trajectories with reproduced robot trajectories. Both paths are resampled to a fixed number of points. Mean trajectory error is calculated as the average Euclidean distance between corresponding sampled points. Path similarity and reproduction accuracy are computed from the normalised trajectory error. Completion success is recorded when the mean trajectory error remains below the success threshold.

Final Evaluation Results

Mean trajectory error increases with artistic trajectory complexity.

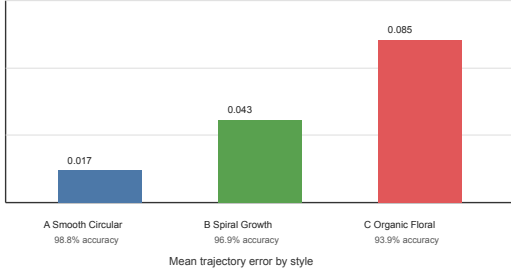


Figure 3: Quantitative evaluation results by style. The graph compares trajectory error and reproduction accuracy across the three artistic movement classes.

7 Results

The final evaluation used five trials per style. Table 1 summarises the numerical results, while Figure 3 provides the main quantitative visual comparison between styles.

Table 1: Final evaluation results by artistic style.

Style	Error	Accuracy	Success	Change
Smooth Circular	0.01684	98.80%	1.00	2.01
Spiral Growth	0.04317	96.91%	1.00	3.40
Organic Floral	0.08526	93.91%	0.40	6.66

7.1 Critical Interpretation

Figure 3 shows a clear performance trend: mean trajectory error increases as the motion style becomes more complex. Style A performs best because its circular structure is regular, periodic, and locally smooth. Once the policy has learned the approximate radius and phase progression, future states remain close to the demonstrated distribution. Style B introduces additional difficulty because the desired radius changes over time; the policy must reproduce both angular movement and outward growth, so small timing errors affect the final trajectory. Style C is hardest because the floral motion has rapid curvature changes and repeated direction reversals. These properties increase compounding error: a small mistake at one part of the petal changes the next observed state, making later predictions less reliable. The results therefore support the hypothesis that Behaviour Cloning is sensitive to trajectory complexity.

Style A achieved the lowest trajectory error and highest reproduction accuracy. Style B produced a higher error because the policy must reproduce a gradually expanding radius over time. Style C produced the highest error and the lowest completion success rate because its floral trajectory has frequent curvature and direction changes. Figure 4 visualises the same trend: the circular reproduction remains closely aligned with the demonstration because the policy repeatedly encounters states similar to the training data. The spiral trajectory diverges slightly because radius growth makes

timing errors visible over the whole path, while the floral trajectory shows larger local deviations because each petal introduces sharper curvature changes. These visual differences are consistent with Behaviour Cloning error accumulation: once the robot departs from the demonstrated state distribution, later actions are predicted from less familiar observations.

8 Discussion

The results support the hypothesis that trajectory complexity affects Behaviour Cloning performance and provide evidence for the brief’s learning objectives: ML algorithm understanding, critical algorithm selection for a simulated robot, and integration of ML-Agents with a robotic manipulation environment. As direction change and curvature variance increased, mean trajectory error also increased. This indicates that the model was less robust when the artistic motion style required more rapid local changes in direction.

The project demonstrates the potential of simulation-based imitation learning for expressive robotic movement. Beyond the numerical trajectory metrics, the work contributes to creative robotics by treating robot motion as an expressive medium rather than only an industrial control problem. Imitation learning is useful for artistic robotic systems because it allows a human demonstrator to define style through movement examples, preserving rhythm and gesture qualities that would be difficult to encode manually as waypoints or hard-coded trajectories. However, Behaviour Cloning learns from observed state-action pairs and does not explicitly model artistic intent or long-term visual composition. It is also vulnerable to covariate shift, where small prediction errors lead the robot into states not represented in the demonstrations.

8.1 Threats to Validity

The findings should be interpreted with several threats to validity in mind. The dataset is limited in size and was generated from a narrow set of demonstrations, so the trained policies may reflect the specific demonstration style rather than a broad distribution of artistic movement. The demonstrations were produced by a single user or scripted reference source, which limits variation in speed, stroke pressure, and stylistic interpretation. Evaluation was conducted only in simulation, so real-world constraints such as brush deformation, paint viscosity, actuator noise, calibration error, and contact dynamics were not measured. The number of evaluation trials is also small, meaning the reported metrics are useful for project comparison but should not be interpreted as statistically general results.

9 Limitations and Future Work

The main limitation is covariate shift. Behaviour Cloning assumes that the learner will encounter states similar to the expert demonstrations. In practice, small prediction errors can move the robot away from the demonstrated trajectory, especially in Style C where local curvature changes quickly. This leads to error accumulation: once the brush deviates from the expected petal path, later actions are predicted from a less familiar state and the visual output becomes less stable.

A second limitation is that the model imitates motion rather than artistic intent. The policy learns how joint actions correspond to

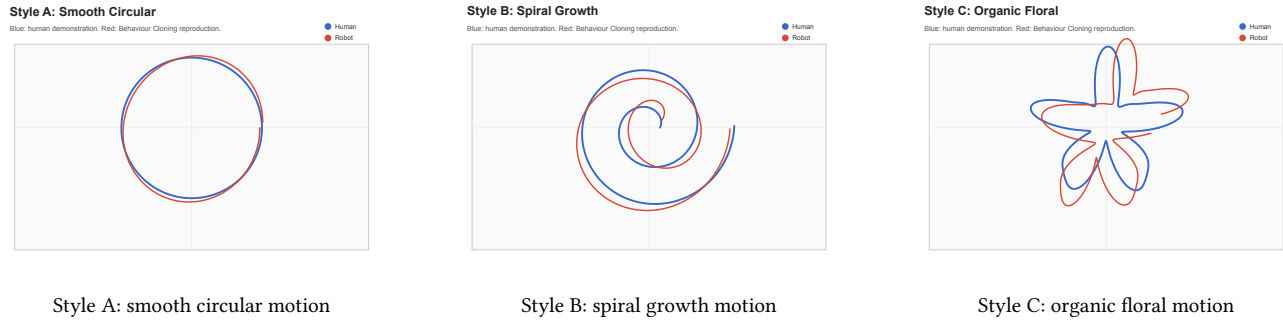


Figure 4: Human versus robot trajectory overlays. Blue trajectories indicate human demonstrations, while red trajectories indicate trained robot reproductions.

demonstrated trajectories, but it does not understand composition, balance, or aesthetic preference. The system also lacks visual perception feedback from the canvas, so it cannot correct its behaviour based on the drawing already produced. Finally, the quality of the learned behaviour depends strongly on the quality and diversity of the demonstrations. More varied demonstrations would likely improve robustness, while noisy or inconsistent demonstrations would reduce imitation accuracy.

Future work could include DAgger-style corrective demonstrations, recurrent policies, inverse kinematics constraints, visual feedback from the canvas, and a larger dataset. A combined style-conditioned model could also be evaluated to test whether one policy can switch between multiple expressive styles.

10 Conclusion

This project asked whether Behaviour Cloning can reproduce expressive artistic brush stroke styles in a simulated robotic painting task. The answer is yes for simple and moderately complex styles: the circular and spiral policies achieved high reproduction accuracy and successful visual outputs. Performance decreased for the more complex floral style, where frequent curvature and direction changes caused larger compounding errors. The project therefore demonstrates the viability of imitation learning for creative robotic motion in simulation, while also showing that trajectory complexity remains a key challenge. Its main contribution is a complete Unity ML-Agents pipeline that connects demonstration collection, Behaviour Cloning, ONNX deployment, visual robotic painting, and quantitative evaluation for expressive movement learning.

11 AI Tools Declaration

Generative AI tools were occasionally used to assist with idea exploration and technical problem-solving during the project. All implementation, experimentation, evaluation, and written content submitted for assessment were completed, verified, and edited by the student.

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